

MALIN HEAD, Ireland

WMO: 039800

Lat: 55.367N

Lon: 7.333W

Elev: 25

StdP: 101.03

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.2	1.4	-4.2	2.7	2.2	-2.9	3.0	3.1	21.8	6.0	20.3	6.6	5.3	190	0.837

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.2	19.9	16.4	18.5	15.7	17.5	15.1	17.1	19.0	16.3	17.9	15.6	17.1	7.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	18.1	15.5	11.1	17.3	14.8	10.6	16.5	48.2	19.1	45.8	18.0	43.7	17.1	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.3	14.5	DB	-1.7	23.2	1.2	1.3	-2.6	24.2	-3.3	25.0	-4.0	25.7	-4.9	26.7	(4)
(5)			WB	-2.6	18.8	1.1	0.7	-3.4	19.3	-4.1	19.8	-4.7	20.2	-5.5	20.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	6.1	6.2	6.9	8.5	10.5	12.7	14.4	14.7	13.5	11.2	8.6	6.8	(6)	
	DBStd	3.73	2.21	2.22	2.23	2.22	2.11	1.86	1.74	1.51	1.73	2.02	2.15	2.43	(7)	
	HDD10.0	566	120	109	97	54	18	1	0	0	0	11	52	103	(8)	
	HDD18.3	3031	378	341	353	294	242	169	122	113	145	222	293	359	(9)	
	CDD10.0	579	1	1	3	11	34	82	137	147	106	47	9	3	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	2	0	0	0	0	1	0	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.5	9.0	8.8	8.2	7.3	6.7	6.2	5.7	6.2	7.1	8.0	8.4	8.7	(14)	
(15) Precipitation	PrecAvg	1096	116	84	83	63	62	70	79	94	96	114	118	114	(15)	
	PrecMax	1402	204	200	152	125	137	141	152	183	216	252	228	273	(16)	
	PrecMin	841	22	10	19	19	15	29	16	13	23	30	25	42	(17)	
	PrecStd	122	46	43	33	25	29	24	31	40	40	48	41	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	12.9	13.9	16.9	19.1	20.5	21.8	21.9	20.2	17.3	14.7	13.6	(19)	
		MCWB	10.8	10.9	10.9	12.2	14.1	15.6	17.4	17.9	16.4	14.9	13.0	12.0	(20)	
	2%	DB	11.1	11.5	12.0	14.3	16.3	18.1	19.7	19.5	18.3	15.9	13.3	12.2	(21)	
		MCWB	9.9	9.9	9.9	11.2	12.9	14.8	16.6	16.6	15.7	14.1	12.0	10.8	(22)	
	5%	DB	10.2	10.5	10.9	12.8	14.7	16.5	18.2	18.1	17.1	15.0	12.4	11.2	(23)	
		MCWB	9.2	9.2	9.2	10.4	12.1	14.0	15.9	15.8	14.9	13.2	11.2	9.9	(24)	
	10%	DB	9.5	9.6	10.1	11.6	13.5	15.4	17.1	17.0	16.1	14.1	11.7	10.3	(25)	
		MCWB	8.4	8.3	8.6	9.5	11.3	13.4	15.0	15.1	14.2	12.6	10.4	9.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.4	11.5	13.3	15.2	16.7	18.2	18.4	17.4	15.3	13.4	12.4	(27)	
		MCDB	12.1	12.4	13.1	15.8	18.0	18.9	20.6	20.7	19.2	16.8	14.3	13.5	(28)	
	2%	WB	10.2	10.2	10.3	11.7	13.5	15.2	17.0	17.2	16.2	14.4	12.3	11.1	(29)	
		MCDB	10.9	11.2	11.5	13.8	15.5	17.4	19.0	18.9	17.7	15.6	13.1	12.1	(30)	
	5%	WB	9.5	9.5	9.6	10.6	12.4	14.4	16.1	16.3	15.4	13.6	11.5	10.2	(31)	
		MCDB	10.2	10.4	10.7	12.4	14.3	16.2	17.9	17.7	16.7	14.8	12.3	11.2	(32)	
	10%	WB	8.6	8.5	8.9	9.8	11.7	13.6	15.4	15.6	14.6	12.8	10.7	9.3	(33)	
		MCDB	9.5	9.6	10.0	11.3	13.4	15.2	16.9	16.8	15.9	14.0	11.6	10.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	4.0	4.3	4.6	4.7	4.3	4.3	4.2	4.2	4.0	3.7	3.8	(35)	
		MCDBR	4.3	4.3	5.2	6.5	6.7	6.3	6.2	6.0	5.7	4.8	4.1	4.5	(36)	
	5% WB	MCWBR	4.3	3.9	4.0	4.3	4.3	4.1	4.1	4.1	4.1	4.0	4.0	4.5	(37)	
		MCWBR	4.2	4.3	4.6	5.9	6.1	5.7	5.7	5.2	5.0	4.6	4.1	4.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.353	0.372	0.382	0.388	0.376	0.379	0.365	0.363	0.349	0.335	0.331	(40)	
		taud	2.202	2.289	2.303	2.324	2.355	2.425	2.423	2.476	2.466	2.410	2.271	2.173	(41)	
	Ebn at Noon		545	708	790	839	855	870	859	851	799	710	559	462	(42)	
		Edh at Noon	61	80	99	111	114	107	106	94	84	70	56	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.50	1.19	2.28	3.81	4.91	5.00	4.64	3.76	2.56	1.41	0.62	0.36	(44)	
	RadStd		0.05	0.10	0.18	0.26	0.43	0.47	0.32	0.30	0.15	0.13	0.06	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page